# Comparing ADMM and Penalty Methods for Distributed Multi-Agent Trajectory Planning

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18-660 / 18-460 Lightning Talk
April 23, 2026

# Background & Motivation

- **Problem.** $N$ drones must reach individual goals while avoiding collisions — but no central coordinator sees all states.
- **Per-agent MPC** (convex QP after linearizing $\|p_i - p_j\| \geq d_{\min}$ at each step):

$$\min_{x_i, u_i} \sum_{k=0}^{H-1} \left( \|x_i(k) - x_i^{\text{ref}}(k)\|_Q^2 + \|u_i(k)\|_R^2 \right)$$

- **Two distributed paradigms** differ in how they handle the *coupling* between agents:
  - *Penalty:* soft cost on violation.
  - *Consensus ADMM:* dual-variable feedback drives feasibility in the limit.

**Why it matters.**

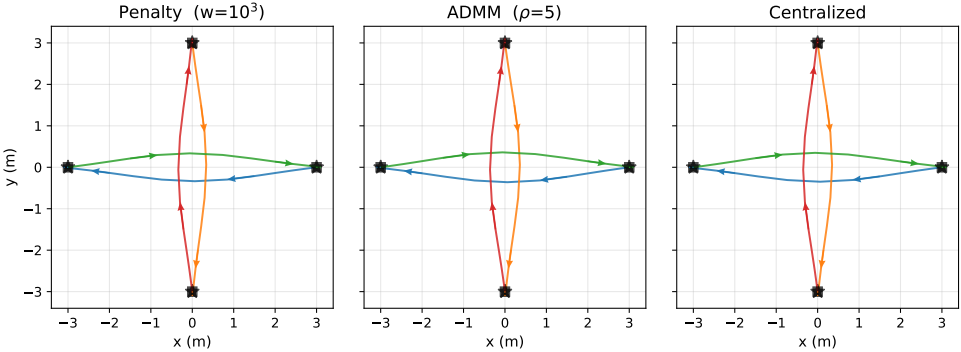
Cyber-physical systems — delivery swarms, robot fleets, multi-vehicle coordination — need distributed optimizers that are both **feasible** (no collisions) and **tractable** (per-agent QPs at each round).

# Key Question

*Does ADMM's dual-variable machinery earn its extra complexity over a simple distributed penalty baseline on multi-drone trajectory planning?*

- Do both methods reach the **centralized optimum**?
- How sensitive is each to its tuning knob — $w$ for penalty, $\rho$ for ADMM?
- Does the comparison hold as we **scale** from $N{=}4$ to $N{=}8$?

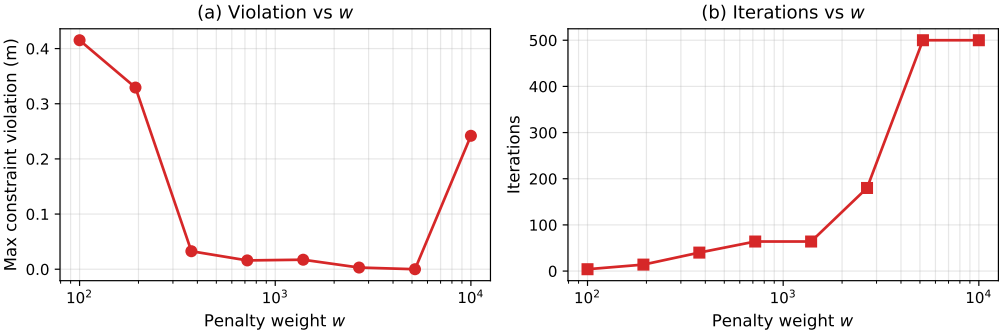
# Take-home 1: All three reach goals; only penalty collides




| S2 ($N$=4, ring graph) | iters | $J^\star$ | min-dist (m) | violation (m) |
|---|---|---|---|---|
| Penalty ($w$=$10^3$) | 99 | 8944.8 | **0.483** | $1.67\times10^{-2}$ |
| ADMM ($\rho$=5) | 44 | 8948.6 | **0.517** | 0 |
| Centralized | 5 | 8946.5 | 0.500 | 0 |

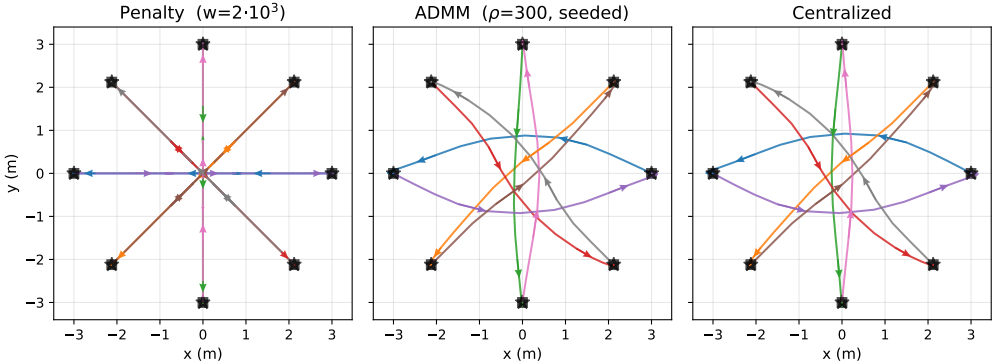
- All three objectives agree within 0.04%, but penalty's **min-distance** 0.483 **m** $< d_{\min}$=0.5 — a real collision hidden by soft slack.

# Take-home 2: Penalty has a U-shape in $w$; ADMM doesn't




- **Small $w$:** collision cost too weak — drones overlap by up to 0.42 m. **Large $w$:** collision cost dominates the QP; the solver can't converge in 500 iterations and still leaves 0.24 m of overlap.
- **Why ADMM has no such knob:** the scaled dual update $\lambda_{ij} \leftarrow \lambda_{ij} + \rho(\theta_i - \theta_j)$ integrates consensus errors iteratively, decoupling feasibility-driving from QP conditioning.

# Take-home 3: Gap widens at $N$=8




| S3 ($N$=8, complete graph) | iters | $J^\star$ | min-dist (m) | violation (m) |
|---|---|---|---|---|
| Penalty ($w$=2·10$^3$) | 20 | **26981** | **0.175** | $3.25\times10^{-1}$ |
| ADMM ($\rho$=300, seeded) | 111 | **17125** | **0.500** | $1.2\times10^{-4}$ |
| Centralized | 5 | 17120 | 0.500 | $\approx 0$ |

- ADMM matches centralized to 0.03% **on** $J^\star$; penalty's drones overlap by 0.33 **m** — the feasibility gap grew ∼20× from S2 to S3.

# When to use which — and why ADMM's duals matter

| | **Penalty** | **ADMM** | **Centralized** |
|---|---|---|---|
| Implementation complexity | low | medium | low |
| Per-iteration cost | $N$ small QPs | $N$ larger QPs | 1 joint QP |
| Feasibility guarantee | none ($w$-dep.) | in the limit | hard |
| Tuning knobs | $w$ | $\rho$ | solver only |
| Work per iteration | $\mathcal{O}(N)$ parallel | $\mathcal{O}(N)$ parallel | $\mathcal{O}(N^2)$ central |

**Thesis.** ADMM's dual-variable feedback pays off exactly where penalty's single $w$ knob runs out of room — the gap grows with the number of coupled agents.

# Thank you

- Code, configs, and animated comparison videos:
  `github.com/chrisjinyu/distributed-trajectory-planning`
- **Selected references.**
  - Boyd, Parikh, Chu, Peleato & Eckstein (2011).
    *Distributed Optimization and Statistical Learning via ADMM.*
  - Chen (2024). *A brief tutorial on consensus ADMM for distributed optimization with applications in robotics.*

Questions?